



Genesis Biotech Inc.

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Pr Taq DNA polymerase (Pfu+Taq)

Catalog No. PG-30003

Quantity: 250 Unit / 125 ul

Storage: -20°C

Description

Pr Taq DNA polymerase is designed for long and accurate amplification of PCR products. It contains both Taq and Pfu DNA polymerase. Taq polymerase is widely used in current PCR practice. A disadvantage of Taq is that it sometimes makes mistakes when copying DNA, leading to mutation in the DNA sequence, since it lacks 3' → 5' proofreading exonuclease activity. Pfu polymerase have proofreading mechanisms and can significantly reduce the number of mutations that occur in the copied DNA sequence. Combinations of both Taq and Pfu are available nowadays that provide both high fidelity and accurate amplification of DNA.

Storage buffer

This enzyme is supplied in 20 mM Tris-HCl, pH 8.0, 100 mM NaCl, 0.1 mM EDTA, 0.5 mM DTT, 1% Triton X-100, 50% glycerol.

10X PCR buffer (1.0ml/vial)

The PCR buffer is supplied as a 10X concentrate and should be diluted for use.

Application

1. Improve PCR efficiency
2. Amplification of longer PCR products
3. Higher fidelity than Taq DNA polymerase
4. Uses less template

Basic PCR Protocol

Add the following components to a sterile 0.5ml microcentrifuge tube sitting on ice:

Components	Volume	Final Concentration
10X PCR buffer	10 ul	1x
(with MgCl ₂)		
2.5mM dNTP	8 ul	200 uM
forward primer	1-5 ul	0.1-0.5 uM
reverse primer	1-5 ul	0.1-0.5 uM
Pr Taq DNA polymerase	0.25-1 ul	0.5-2 unit
(2U/ul)		
Template < 0.5 ug	0.05-0.5 ug for genomic DNA	
	0.5-50 ng for plasmid	

Sterile distilled water up to 100 ul.

Unit definition

One unit is the amount of enzyme required to catalyze the incorporation of 10 nmol of dNTP into acid-insoluble material in 30 minutes at 74°C, pH9.3, with activated salmon sperm DNA as the template-primer. The reaction conditions are: 25 mM TAPS (pH 9.3 at 25°C), 1 mM 2-mercaptoethanol, 2 mM MgCl₂, 50 mM KCl, 200 uM each of dATP, dTTP, dGTP, 100 uM [α-³²P]dCTP, 0.25 mg/ml activated salmon sperm DNA.

Purity

Nicking activity, endonuclease and exonuclease activity were not detected after the incubation of 0.6 ug of supercoiled pUC18 DNA, 0.6 ug of Lambda DNA or 0.6ug of Lambda-HindIII digest with 10 units of this enzyme for 1 hour at 74°C.

Related products

Taq DNA polymerase	PG-30001
Hotstart Taq DNA polymerase	PG-30002
Hotstart Pr Taq DNA polymerase (Taq+Pfu)	PG-30004
Klen Taq DNA polymerase	PG-30005
Hotstart Klen Taq DNA polymerase	PG-30006
Anti-Taq Hotstart Antibody	PG-30007
Real-time PCR kit	PG-30009
One Step RT-PCR kit	PG-30010
Genotyping kit	PG-30012
PCR kit	PG-30013

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